Leamer Slides

April 19, 2022

The Con in Econometrics Poses a Hypothetical Additional Data Set

- A Farmer notices that the yields are greater under the trees where the bird-droppings land, and thinks the cause is the bird-droppings.
- A Neighbor argues that actually it's the shade under the tree that affects the yield.
- The data are completely silent on which is right but the Farmer is very doubtful that shade matters at all, and the Neighbor is very sure that bird droppings could not be the cause.
- Each walks away sure of their own distinct conclusion from the data.
- Message: When the data are weak it takes prior opinion to allow the data to be interpreted. That opinion can be summarized by a fictitious prior data set which is pooled with the actual data. Different priors lead to different inferences from the same data set.

Warning:

- The set of parameters created by lists of multiple predictors, interactions, nonlinearities and different lag structures overwhelms any economic data. This is true even in experimental settings.
- The data need help to be useful. When given the task of estimating all the parameters, the data replies: I cannot do that. If you can pick some, then I can pick the others.
- The usual worthless high-dimensional real data can be turned into something useful by creating a fictitious prior data set that captures the analyst's opinions about the context. The data set is useful if the pooled results are very different from the prior results.
- **The con in Econometrics is to use prior information in each and every setting but to pretend it plays no role and to claim: "THAT RESIDUAL IN MY FINAL MODEL IS A RANDOM VARIABLE CHOSEN BY NATURE "**

Question: Will a Rise in Short-Term Interest Rates Reduce the Growth Rate of Real GDP in 2022 and 2023?

- The dependent variable is the growth rate of real GDP, expressed as dlog(GDPC1) using the FRED language for real GDP.

Dependent Variable: DLOG(GDPC1)				
Method: Least Squares				
Date: 04/18/22 Time: 06:08				
Sample: 1986Q1 2020Q1				
Included observations: 137				
Variable	Coefficien	Std. Error	t-Statistic	Prob.
C	0.00374	0.001	4.20	0
DLOG(GDPC1(-1))	0.28383	0.090	3.17	0.0019
TB3MS	0.00026	0.000	1.34	0.182
TB3MS-TB3MS(-4)	0.00088	0.000	2.25	0.0259
R-squared	0.18	Mean dependent var		0.0062
Adjusted R-squared	0.16	S.D. dependent var		0.0058
S.E. of regression	0.01	Akaike info criterion		-7.5959

The Prosecution Claims:
This proves higher interest rates come with higher rates of growth. Good news for Jerome Powell!!

The Defense Replies:
Not if you include other variables!


Reasonable Doubt

Important Theorem: t-statistic measures sturdiness

- If you omit a variable, the coefficient for a variable with an absolute t-value larger than the omitted variable cannot change sign.
- If you want to change the signs: omit variables with big t-values.
- This is a big step toward the conclusion that **absolute t-values measure statistical significance of the estimate and relative t-values measure sturdiness of the estimate**.

An Overloaded Equation

Explaining the Growth
Of Real GDP

	Dependent Variable: DLOG(GDPC1)				
	Method: Least Squares				
	Date: 04/17/22 Time: 09:29				
	Sample (adjusted): 1986Q1 2020Q1				
	Included observations: 137 after adjustments				
	Variable	Coefficien	Std. Error	t-Statistic	Prob.
	C	0.0070	0.0026	2.74	0.007
	DLOG(GDPC1(-1))	-0.2205	0.0938	**-2.35**	0.021
3-month yield	TB3MS	-0.0006	0.0007	-0.91	0.366
	TB3MS-TB3MS(-4)	0.0004	0.0005	0.71	0.477
10-year yield	GS10	0.0007	0.0008	0.97	0.333
	GS10-GS10(-4)	-0.0002	0.0006	-0.36	0.717
Unemployment rate	UNRATE	-0.0009	0.0004	-1.96	0.052
	UNRATE-UNRATE(-4)	0.0019	0.0018	1.06	0.290
Inflation	DLOG(CPIAUCSL)	0.0031	0.1044	0.03	0.976
	LOG(CPIAUCSL/CPIAUCSL(-4))	-0.0645	0.0512	-1.26	0.210
Working Age Population	DLOG(LFWA64TTUSM647S)	0.0500	0.2770	0.18	0.857
	LOG(LFWA64TTUSM647S/LFWA64TTUSM647S(-4))	0.1396	0.1287	1.08	0.280
Male Labor Force Participation Rate	DLOG(LNS11300001)	-0.0166	0.1702	-0.10	0.923
	LOG(LNS11300001/LNS11300001(-4))	-0.0230	0.1272	-0.18	0.857
Female Labor Force Participation Rate	DLOG(LNS11300002)	-0.1400	0.2017	-0.69	0.489
	DLOG(LNS11300002/LNS11300002(-4))	0.0773	0.1317	0.59	0.559
Payroll Jobs	**DLOG(PAYEMS)**	1.8181	0.2738	**6.64**	0.000
	LOG(PAYEMS/PAYEMS(-4))	-0.1855	0.1139	-1.63	0.106
Imports from China	**DLOG(IMPCH)**	0.0056	0.0026	**2.13**	0.035
	LOG(IMPCH/IMPCH(-4))	0.0060	0.0053	1.11	0.268
	R-squared	0.55	Mean dependent v		0.00622
	Adjusted R-squared	0.47	S.D. dependent va		0.00585
	S.E. of regression	0.00	Akaike info criteri		-7.950

How can we reduce the parameterization and concentrate the limited data resource on a smaller set of questions?

- Stepwise Regression: Sequentially omit the variable with the lowest t.
  - This is a possible algorithm but it leads to dishonest data summaries, sometimes called p-hacking.
- Bayesian analysis which combines the actual data set with a fictitious data set designed to capture the analyst’s understanding of the context.
  - Include a sensitivity analysis that shows how much the conclusions change as the fictitious data set changes
  - This is honesty. It’s not unscientific as most of my colleagues assert. Unscientific is not to disclose how the parameterization was reduced and pretend that your results are 100% reliable. They call that scientific. That’s the con.

Stepwise Regression

Omissions Until All Coefficients Have absolute t-values greater than one.

	Dependent Variable: DLOG(GDPC1)				
	Method: Least Squares				
	Date: 04/17/22 Time: 11:12				
	Sample (adjusted): 1986Q1 2020Q1				
	Included observations: 137 after adjustments				
	Variable	Coefficier	Std. Error	t-Statistic	Prob.
	C	0.0080	0.0023	3.41	0.001
	DLOG(GDPC1(-1))	-0.1918	0.0876	**-2.19**	0.030
3-month yield					
10-year yield					
Unemployment rate	UNRATE	-0.0006	0.0003	-2.05	0.043
	UNRATE-UNRATE(-4)	0.0017	0.0010	1.68	0.096
Inflation					
	LOG(CPIAUCSL/CPIAUCSL(-4))	-0.0693	0.0366	**-1.89**	0.061
Working Age Population	DLOG(LFWA64TTUSM647S)	0.2239	0.2103	1.06	0.289
Male Labor Force Participation Rate					
Female Labor Force Participation Rate					
Payroll Jobs	DLOG(PAYEMS)	1.7967	0.2285	**7.86**	0.000
	LOG(PAYEMS/PAYEMS(-4))	-0.1986	0.0747	-2.66	0.009
Imports from China	DLOG(IMPCH)	0.0059	0.0025	**2.35**	0.020
	LOG(IMPCH/IMPCH(-4))	0.0091	0.0038	2.38	0.019
	R-squared	0.532	Mean dependent v		0.00622
	Adjusted R-squared	0.499	S.D. dependent va		0.006
	S.E. of regression	0.0041	Akaike info criteri		-8.067

Bounds for the Free Coefficients When Doubtful Variables Can Be Played With: Study 1

Free Variables

Doubtful Variables

	Dependent Variable: DLOG(GDPC1)				
	Method: Least Squares				
	Date: 04/17/22 Time: 11:09				
	Sample (adjusted): 1986Q1 2020Q1				
	Included observations: 137 after adjustments				
	Variable	Coefficien	Std. Error	t-Statistic	Prob.
	C	0.0070	0.0026	2.75	0.007
	DLOG(GDPC1(-1))	-0.2176	0.0937	**-2.32**	0.022
3-month yield	TB3MS	-0.0006	0.0007	-0.83	0.411
	TB3MS-TB3MS(-4)	0.0003	0.0006	0.63	0.529
10-year yield	GS10	0.0006	0.0008	0.70	0.482
	GS10-GS10(-4)	-0.0002	0.0006	-0.26	0.796
Unemployment rate	UNRATE	-0.0008	0.0004	-1.80	0.075
	UNRATE-UNRATE(-4)	0.0017	0.0019	0.91	0.367
Inflation	DLOG(CPIAUCSL)	0.0038	0.1044	0.04	0.971
	LOG(CPIAUCSL/CPIAUCSL(-4))	-0.0588	0.0516	-1.14	0.257
Working Age Population	DLOG(LFWA64TTUSM647S)	0.0537	0.2770	0.19	0.847
	LOG(LFWA64TTUSM647S/LFWA64TTUSM647S(-4))	0.1628	0.1349	1.21	0.230
Male Labor Force Participation Rate	DLOG(LNS11300001)	-0.0102	0.1698	-0.06	0.952
	LOG(LNS11300001/LNS11300001(-4))	-0.0390	0.1281	-0.30	0.762
Female Labor Force Participation Rate	DLOG(LNS11300002)	-0.0820	0.1535	-0.53	0.594
	LOG(LNS11300002/LNS11300002(-4))	0.0403	0.0938	0.43	0.668
Payroll Jobs	**DLOG(PAYEMS)**	1.8116	0.2735	**6.62**	0.000
	LOG(PAYEMS/PAYEMS(-4))	-0.1992	0.1185	-1.68	0.096
Imports from China	**DLOG(IMPCH)**	0.0057	0.0026	**2.18**	0.031
	LOG(IMPCH/IMPCH(-4))	0.0060	0.0054	1.12	0.265
	R-squared	0.54	Mean dependent var		0.00622
	Adjusted R-squared	0.47	S.D. dependent var		0.00585
	S.E. of regression	0.00	Akaike info criterion		-7.948

Extreme Bounds		
Lower	upper	**s-value**
-0.260	0.327	**0.11**
-0.003	0.003	**-0.05**
-0.002	0.003	**0.28**

An s-value is the average of the upper and lower divided by half the difference. Just like a t-value,

Bounds For the Free Coefficients When A Different Set of Doubtful Variables Can Be Played With: Study 2

	Dependent Variable: DLOG(GDPC1)				
	Method: Least Squares				
	Date: 04/17/22 Time: 11:28				
	Sample (adjusted): 1986Q1 2020Q1				
	Included observations: 137 after adjustments				
	Variable	Coefficier	Std. Error	t-Statistic	Prob.
	C	0.0070	0.0026	2.75	0.007
	DLOG(GDPC1(-1))	-0.2176	0.0937	**-2.32**	0.022
3-month yield	TB3MS	-0.0006	0.0007	-0.83	0.411
	TB3MS-TB3MS(-4)	0.0003	0.0006	0.63	0.529
Payroll Jobs	**DLOG(PAYEMS)**	1.8116	0.2735	**6.62**	0.000
	LOG(PAYEMS/PAYEMS(-4))	-0.1992	0.1185	-1.68	0.096
10-year yield	GS10	0.0006	0.0008	0.70	0.482
	GS10-GS10(-4)	-0.0002	0.0006	-0.26	0.796
Unemployment rate	UNRATE	-0.0008	0.0004	-1.80	0.075
	UNRATE-UNRATE(-4)	0.0017	0.0019	0.91	0.367
Inflation	DLOG(CPIAUCSL)	0.0038	0.1044	0.04	0.971
	LOG(CPIAUCSL/CPIAUCSL(-4))	-0.0588	0.0516	**-1.14**	0.257
Working Age Population	DLOG(LFWA64TTUSM647S)	0.0537	0.2770	0.19	0.847
	LOG(LFWA64TTUSM647S/LFWA64TTUSM64	0.1628	0.1349	1.21	0.230
Male Labor Force Participation Rate	DLOG(LNS11300001)	-0.0102	0.1698	-0.06	0.952
	LOG(LNS11300001/LNS11300001(-4))	-0.0390	0.1281	-0.30	0.762
Female Labor Force Participation Rate	DLOG(LNS11300002)	-0.0820	0.1535	-0.53	0.594
	LOG(LNS11300002/LNS11300002(-4))	0.0403	0.0938	0.43	0.668
Imports from China	**DLOG(IMPCH)**	0.0057	0.0026	**2.18**	0.031
	LOG(IMPCH/IMPCH(-4))	0.0060	0.0054	1.12	0.265
	R-squared	0.545	Mean dependent var		0.00622
	Adjusted R-squared	0.471	S.D. dependent var		0.00585
	S.E. of regression	0.004	Akaike info criterion		-7.948

Free Variables

Doubtful Variables

Extreme Bounds

Lower	upper	**s-value**
-0.241	-0.078	-1.97
-0.002	0.002	-0.04
-0.001	0.001	0.24
1.391	2.189	4.49
-0.494	-0.001	-1.00

Other issues

- What if the effect of the interest rate works through the unemployment rate: an inverted yield curve causes a recession with elevated unemployment and lower GDP growth?
- What if it’s the slope of the yield curve that determines when a recession will occur? How can the analysis be altered to allow for this?

Become a Billionaire with Me

- Artificial intelligence/machine learning is like stepwise regression. It is usually ignorant of the context and it ignores the sensitivity issues.
- Join me in offering an alternative to artificial intelligence that begins with a conversation between analyst and computer that is intended to capture the analyst's view of the context such as which variables are free and which are doubtful. It also informs the analyst what is needed to conclude with confidence that a coefficient is positive, allowing for both statistical significance and also sturdiness.

Key takeaway:
The context matters

Leamer articles on these topics

- "A Result on the Sign of Restricted Least Squares Estimates," **Journal of Econometrics**, 3 (1975), 387-390.
- "Matrix Weighted Averages and Posterior Bounds," with G. Chamberlain, **Journal of the Royal Statistical Society**, Series B, 38 (No. 1 1976), 73-84.
- **Specification Searches: Ad Hoc Inference with Non Experimental Data**, John Wiley and Sons, Inc., 1978. Translated into Spanish and Russian.
- "Sets of Posterior Means with Bounded Variance Priors," **Econometrica**, 50 (May 1982), 725-736.
- “S-values: Conventional context-free measures of the sturdiness of regression coefficients,” **Journal of Econometrics,** Volume 193, Issue 1, July 2016, 147–161.